Batch: A2	Roll No.: 16010324044
Experiment / assignment / tutorial No. 3	
Grade: AA / AB / BB / BC / CC / CD /DD	
Signature of the Staff In-charge with date	

TITLE: Configuration of a CISCO Router.

AIM: To configure a router for given network topology

OUTCOME: Student will be able to understand layered architecture of communication and the configuration of a router for the given topology

Theory:

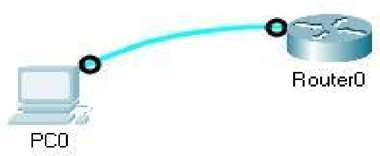
A router is a networking device that forwards data packets between computer networks. Routers perform the traffic directing functions on the Internet. Data sent through the internet, such as a web page or email, is in the form of data packets. A packet is typically forwarded from one router to another router through the networks that constitute an internetwork (e.g. the Internet) until it reaches its destination node.

A router is connected to two or more data lines from different IP networks. When a data packet comes in on one of the lines, the router reads the network address information in the packet header to determine the ultimate destination. Then, using information in its routing table or routing policy, it directs the packet to the next network on its journey.

The most familiar type of IP routers are home and small office routers that simply forward IP packets between the home computers and the Internet. More sophisticated routers, such as enterprise routers, connect large business or ISP networks up to the powerful core routers that forward data at high speed along the optical fiber lines of the Internet backbone.

Procedure:

Part 1: Verify the Default Router Configuration



Select ISR 4331 router

Step 1: Establish a console connection to R1.
a. Choose a **Console** cable from the available connections.

b. Click **PCA** and select **RS 232**.

c. Click **R1** and select **Console**.

d. Click **PCA** > **Desktop** tab > **Terminal**.

e. Click **OK** and press **ENTER**. You are now able to configure **R1**

Step 2: Enter privileged mode and examine the current configuration
You can access all the router commands from privileged EXEC mode. However, because many of the privileged commands configure operating parameters, privileged access should be password-protected to prevent unauthorized use.

a. Enter privileged EXEC mode by entering the **enable** command.
Router> **enable**
Router#
Notice that the prompt changed in the configuration to reflect privileged EXEC mode.

b. . Enter the **show ip interface brief** command.

Router# **show ip interface brief**

What is the router's hostname? **Router**

How many Fast Ethernet interfaces does the Router have? **0**

How many Gigabit Ethernet interfaces does the Router have? **3**

How many Serial interfaces does the router have? **0**

What is the range of values shown for the vty lines? **0-4**

Router# **show running-config**

c. Display the current contents of NVRAM.

Router# **show ip interface brief**

Part 2: Configure and Verify the Initial Router Configuration
To configure parameters on a router, you may be required to move between various configuration modes. Notice how the prompt changes as you navigate through the IOS configuration modes.

Step 1: Configure the initial settings on R1.

a. Configure **R1** as the hostname.

b. Configure Message of the day text: **Unauthorized access is strictly prohibited.**

c. Encrypt all plain text passwords

Use the following passwords:

1) Privileged EXEC, unencrypted: **abcd**
2) Privileged EXEC, encrypted: **itsasecret**
3) Console: **letmein**

Step 2: Verify the initial settings on R1.

a. Verify the initial settings by viewing the configuration for R1.

What command do you use? **show running-config**

b. Exit the current console session until you see the following message:

R1 con0 is now available

Press RETURN to get started

c. Press **ENTER**; you should see the following message:

Unauthorized access is strictly prohibited.
User Access Verification
Password:

Why should every router have a message-of-the-day (MOTD) banner?
It displays a legal warning to unauthorized users and informs them that access is restricted. It also discourages unauthorized login attempts.

If you are not prompted for a password before reaching the user EXEC prompt, what console line command did you forget to configure? **login**

d. Enter the passwords necessary to return to privileged EXEC mode

If you configure any more passwords on the router, are they displayed in the configuration file as plain text or in encrypted form? Explain **plain text, if service password-encryption cmd used then encrypted form**

Part 3: Save the Running Configuration File
Step 1: Save the configuration file to NVRAM.
a. You have configured the initial settings for **R1**. Now back up the running configuration file to NVRAM to ensure that the changes made are not lost if the system is rebooted or loses power

What command did you enter to save the configuration to NVRAM?
copy running-config startup-config

Step 2: Optional: Save the startup configuration file to flash.
Although you will be learning more about managing the flash storage in a router in later chapters, you may be interested to know that, as an added backup procedure, you can save your startup configuration file to flash. By default, the router still loads the startup configuration from NVRAM, but if NVRAM becomes corrupt, you can restore the startup configuration by copying it over from flash.
Complete the following steps to save the startup configuration to flash.

a. Examine the contents of flash using the **show flash** command:

R1# show flash

How many files are currently stored in flash?
The largest file with a .bin extension is the Cisco IOS image.

Which of these files would you guess is the IOS image?
c1900-universalk9-mz.SPA.152-4.M3.bin

b. Use the **show flash** command to verify the startup configuration file is now stored in flash.

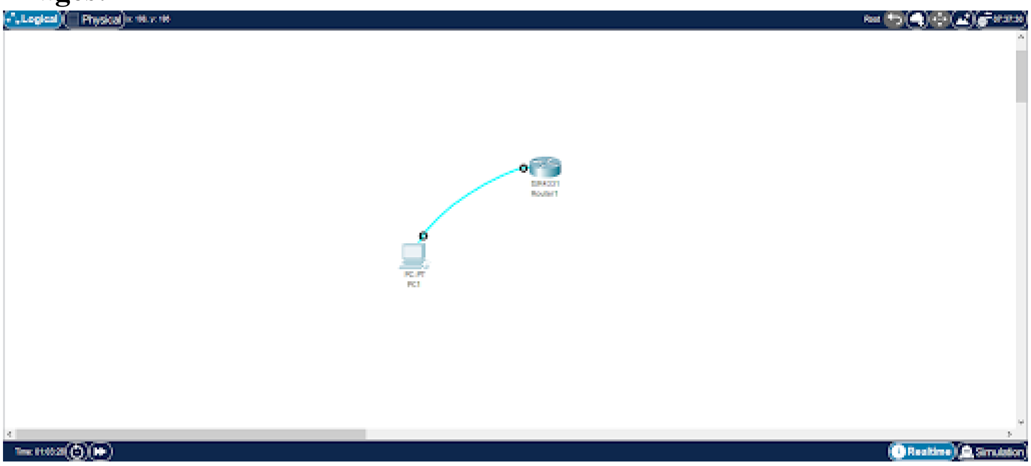
Why do you think this file is the IOS image?
Because it is the largest executable .bin file stored in flash memory and contains the operating system that the router loads during boot.

R1# **copy startup-config flash**

Destination filename [startup-config]

The router prompts you to store the file in flash using the name in brackets. If the answer is yes, then press **ENTER**; if not, type an appropriate name and press **ENTER**.

Images:



```
Router>enable
Router#confiure terminal
                ^
% Invalid input detected at '^' marker.

Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#username Dhairya password abcd
R1(config)#line console 0
R1(config-line)#login local
R1(config-line)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#enable secret itsasecret
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#line vty 0 15
R1(config-line)#password letmein
R1(config-line)#login
R1(config-line)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#
```

```
Username: vedantsalimath
Password:
vedantsalimath>
*Jul 30 10:11:55.373: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: vedantsalimath] [Source: UNKNOWN] [localport: 0] at 10:11:55 UTC Thu Jul 30 2026
vedantsalimath>enable
Password:
vedantsalimath#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
vedantsalimath(config)#hostname Dhairya
Dhairya(config)#exit
Dhairya#
*Jul 30 10:12:55.871: %SYS-5-CONFIG_I: Configured from console by vedantsalimath on console
Dhairya#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Dhairya(config)#username cisco password abcd
 WARNING: Command has been added to the configuration using a type 0 password. However, type 0 passwords will soon be deprecated. Migrate to a supported pass
Dhairya(config)#
*Jul 30 10:16:32.103: %AAAA-4-CLI_DEPRECATED: WARNING: Command has been added to the configuration using a type 0 password. However, type 0 passwords will so
 password type
Dhairya(config)#line console 0
Dhairya(config-line)#login local
Dhairya(config-line)#end
Dhairya#
*Jul 30 10:16:56.118: %SYS-5-CONFIG_I: Configured from console by vedantsalimath on console
Dhairya#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Dhairya(config)#enable secret itsasecret
Dhairya(config)#exit
Dhairya#
*Jul 30 10:17:45.006: %SYS-5-CONFIG_I: Configured from console by vedantsalimath on console
Dhairya#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Dhairya(config)#line vty 0 15
Dhairya(config-line)#password letmein
Dhairya(config-line)#login
Dhairya(config-line)#end
Dhairya#
*Jul 30 10:18:27.294: %SYS-5-CONFIG_I: Configured from console by vedantsalimath on console
Dhairya#
```

CONCLUSION

In this experiment, I successfully verified the default router configuration, configured the hostname, passwords, and MOTD banner, encrypted passwords, verified the configuration, and saved the running configuration to NVRAM and flash memory. The experiment demonstrated the basic configuration and management of a Cisco ISR4331 router.

Signature of faculty in-charge